PPE Compliance Detection — Accuracy Checklist

Status as of 30 Oct 2026 • Project folder: `D:\software\ppe-detection` • Focus: accuracy

1. Where we are right now

> **The MVP works end to end.** Upload a photo → each worker is detected → helmet and vest are matched to that worker → a verdict per worker (OK / violation / needs review) → annotated image, table, summary and CSV/JSON download. All 48 automated tests pass. The web app runs at `http://localhost:8501`.

> **What is not done yet is proving and raising accuracy on your own hard cases.** The numbers below come from the dataset's own test split, which is the easy case. The hard cases from the problem statement (helmet in hand, helmet on a table, hi-vis clothing that is not a vest, dark and distant workers) have not been measured because there are no test photos yet.

2. Accuracy scoreboard (measured, not estimated)

Final model: YOLO11s fine-tuned for 60 epochs (36 minutes). Scored on the **141-image test split**, which was never used for training or for choosing the checkpoint. Detection confidence 0.25.

Class	Precision	Recall	mAP@50	Plain meaning
All	0.890	0.862	0.866	Roughly 9 in 10 detections are right, and about 86% of real items are found
Person	0.850	0.818	0.809	**Weakest.** A missed person means no verdict for that worker
Helmet	0.936	0.919	0.920	Strongest
Vest	0.883	0.848	0.870	Good, but untested on look-alike clothing

Inference size: 960 px beat 1280 px (mAP@50 0.866 vs 0.854; person dropped from 0.809 to 0.790 at 1280). So 960 stays the default. I had expected a larger size to help distant workers; this test did not support that.

> **Read these numbers with care.** The dataset has only about 500 distinct scenes, many of them near-identical frames of the same video, so the training, validation and test splits probably share very similar images. These scores are likely higher than what you would get on photos from a different site. Your own hard-case photos are the fair test, and Step B below will check how much overlap there is.

3. Checklist

Completed

Item	Result
✅ Python 3.12 environment with CUDA PyTorch	GPU recognised (RTX 5060)
✅ Dataset prepared (person, helmet, vest only)	1,416 images: 1132 train / 143 val / 141 test
✅ Detector trained (60 epochs)	Test mAP@50 0.866; weights in `weights/best.pt`

✅ Per-worker matching (head zone, torso zone, one-to-one)	A helmet on a table or in a hand is not credited; one helmet cannot clear two workers (unit-tested)
✅ Rules with three outcomes and a reason sentence	COMPLIANT / VIOLATION / NEEDS_REVIEW for small, cut-off or crowded workers
✅ Low-light enhancement (CLAHE)	Applied automatically to dark images
✅ Annotated images, readable in crowds	Green / red / amber boxes; labels no longer overlap
✅ Violation summary, CSV and JSON export	Counts, breakdown by missing item, unassigned items
✅ Command-line run and Streamlit web app	Upload → result in seconds
✅ Evaluation script and ground-truth template	`eval/run_eval.py`, scored per hard-case group; ready, waiting for photos
✅ Honest measurement on the held-out test split	Table in section 2

Remaining — accuracy work, in priority order

Who	Task	Why it improves accuracy	Time
You	☐ **A. Collect 15–20 hard-case photos** into `eval/test_images/`: helmet in hand, helmet on a table, orange or yellow non-vest clothing, 3–4 workers close together, a worker far away, a dark photo, plus a fully equipped and a bare worker. Staged photos are fine; we say so in the README.	Without these, we cannot measure the cases the problem statement cares about. **Everything below depends on this.**	30–40 min
You	☐ **A2. Write what is true for each photo** (left to right: helmet yes/no, vest yes/no), or tell me and I write `eval/ground_truth.csv`.	The answer key for scoring	15 min
Me	☐ **B. Baseline run** of `eval/run_eval.py` and a check for near-duplicate images across train / val / test.	Gives the real starting number and shows how optimistic section 2 is	10 min
You	☐ **C. Download SH17 from Kaggle** (search "SH17 dataset PPE", free login) and unzip to `data/raw/sh17/`. CC BY-NC-SA 4.0: fine for education, credit required.	8,099 industrial images vs our ~500 distinct scenes: the biggest gain in variety	10–15 min
Me	☐ **D. Merge SH17 with the current data and retrain**, then re-score on the same test photos. Keep the new model only if it is better.	More varied scenes, more workers without PPE, fewer false vests and helmets	~1 hr GPU (laptop gets hot; or Kaggle GPU)
You	☐ **E. Hard-negative photos:** 30–50 photos of workers in orange or yellow clothing that is not a vest.	Teaches the model that such clothing is not a vest (this failed on the rafting photo)	20 min
Me	☐ **E2. Label them semi-automatically:** the model drafts the person boxes, you check them, the jacket stays unlabeled; then retrain.	Fixes the look-alike clothing error	40 min + training

Me	☐ **F. Tune thresholds** (confidence per class, zone sizes, review rules) on the hard-case set, one change at a time, keeping a change only if the score improves.	Cuts wrong VIOLATION calls and wrong REVIEW calls without retraining	40 min
Me	☐ **G. Fix the weakest class (person, mAP@50 0.81):** more data from SH17, and if needed a larger model (YOLO11m).	A missed person means a worker is never judged	1 hr GPU
Me	☐ **H. Handle "helmet found but no person found"** (seen on the back-view photo): label it as unassigned, and optionally flag the image for review.	Avoids silently missing a worker	20 min

Remaining — finishing work

Who	Task	Time
Me	☐ Optional video mode: tracking per worker, report a violation only if it persists over several frames	1 hr
Me	☐ README: how it works, setup, results table from the evaluation, known limitations, credits (dataset, libraries, ideas studied, AI assistance)	40 min
Me	☐ 3-minute demo script following the app, including one honest failure case	20 min
Me	☐ Final check: fresh environment, follow the README, run tests, evaluation and the app	30 min
You	☐ Read the hackathon rules for limits on pre-existing code or AI assistance; decide when to set up git (nothing is set up yet, as you asked)	10 min
You	☐ Send the repo links you mentioned, if you still want them read for ideas	—

4. Next 60 minutes (what gets the most accuracy fastest)

1. **You:** start collecting the hard-case photos (A) and start the SH17 download (C). Both can happen at the same time.
2. **Me, meanwhile:** fix the "helmet but no person" case (H) and prepare the merge script for SH17 (D), so retraining can start the minute the files arrive.
3. **Together:** baseline score on your photos (B), then retrain (D). Compare before and after in the same table.

5. Honest limits to keep in mind

- The test numbers above come from one public dataset with many near-duplicate frames. Expect lower numbers on your own photos until more data is added.
- No accuracy figure for the hard cases exists yet. We will report the per-group table from the evaluation script, including failures, and say the thresholds were tuned on the same small set.
- Retraining heats the laptop (about 40 minutes at full GPU load). Kaggle's free GPU is an option if that is a problem.

- The README will credit the dataset, Ultralytics YOLO and the other libraries, list the repos we studied for ideas, and note that Claude was used as a coding assistant.